

# Genomic exon-span states and observation constraints

One ordered physical exon geometry per state

## A Physical exon geometry



Ordered physical intervals  
including terminal and UTR spans

## B Declared DNA-tract states



**One origin: root or one branch**

Absent: 0 or 2; present: 1

## C Tip compatibility

Unknown exon call; DNA present



$m=1$   
 $w=1$



$m=1$   
 $w=1$



$m=0/2$   
 $w=0$

Compatible geometries get weight 1  
Known DNA constraints remain active

## D Finite geometry catalogue

Observed boundaries + tract cuts

## Finite catalogue

Complete enumeration



**No default count cap**

Limits stop; no truncation